

Maryam Torkashvand

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PROFILE

- GIS and spatial data science researcher with 4+ years of experience building Python and R workflows for large-scale spatial networks, mobility analysis, geocoding, and cartographic visualization.
- Proficient in PostgreSQL/PostGIS, NetworkX, ArcGIS Pro, and Survey123, with a track record of peer-reviewed publications in GIS and social science journals.
- PhD candidate in Geographic Information Science at the University of Iowa, with prior work on mobility networks, transportation cartography, and survey-based geospatial outreach.

SKILLS

- **Programming and Data Analysis:** Python (Geopandas, NetworkX, Scikit-learn, Matplotlib, Arcpy), R (tidyverse, sf, ggplot2), SQL (PostgreSQL/PostGIS), JavaScript (D3.js).
- **GIS and Spatial Analysis:** ArcGIS Pro, ArcGIS Online, StoryMaps, Survey123, QGIS, spatial network analysis, geocoding, cartography, raster and vector analysis.
- **Methods:** Spatial statistics, network analysis, machine learning, demographic microsimulation, survey-data collection and processing, data engineering, workflow documentation.
- **Academic Writing:** Peer-reviewed publications in Journal of Hydrology, Historical Methods, The History of the Family, and Transactions in GIS (in review); conference papers and research proposals.

EXPERIENCE

Graduate Research Fellow, Graduate College, University of Iowa

Feb 2026 to Present

Iowa City, IA

- Lead a spatial microsimulation project modeling population and kinship networks across the United States from 1850 to 1920, using large-scale family-tree data, demographic indices from census records, and R.
- Construct kinship networks and quantify their structural and connectivity properties; analyze data biases and validate the impact of missing records.
- Draft a peer-reviewed manuscript documenting the model framework, spatial methods, and preliminary findings.

Graduate Research Assistant (Spatial Data Specialist), School of Earth, Environment, and Sustainability, University of Iowa (NSF-funded project)

May 2023 to Jan 2026

Iowa City, IA

- Built repeatable Python and PostgreSQL/PostGIS pipelines to clean, standardize, and geocode 280M historical place records into analysis-ready county and city outputs with validation summaries.
- Produced publication-quality maps and visualizations of mobility patterns and spatial networks for peer-reviewed journals, proposals, and international conferences.
- Integrated and validated multi-source spatial reference datasets and modified spatial algorithms to produce defensible fine-resolution geocoding results.
- Improved a 60M-node spatial network through rule-based deduplication and data imputation, recovering 10M connections for downstream spatial modeling.
- Wrote manuscripts and presented at GIScience, SSHA, AAG, and CaGIS conferences.

Graduate Research Associate, IIHR Hydrosience and Engineering, University of Iowa (NSF-funded project)

Iowa City, IA

May 2024 to Aug 2024

- Built the "[Rivers and Risks](#)" StoryMap in ArcGIS Online and designed Survey123 instruments for stakeholder outreach and structured survey-data collection.

- Created an animated map of Iowa population growth and transportation development using JavaScript and D3.js.
- Translated engineering water-quality risk assessments into maps, visualizations, and narratives for non-technical audiences.

Teaching Assistant, Department of Geographical and Sustainability Sciences, University of Iowa

Iowa City, IA

Aug 2022 to May 2023

- Taught two hands-on GIS laboratory sections covering ArcGIS Pro, spatial-analysis workflows, and cartographic outputs; mentored 42+ students on vector and raster GIS projects.

EDUCATION

Ph.D., Geography (GIScience and Cartography), University of Iowa

Aug 2022 to Dec 2026 (expected)

M.Sc., Remote Sensing and GIS (Satellite Meteorology), Islamic Azad University, Tehran

2017 to 2020

B.E., Surveying Engineering, Shahid Rajaei Teacher Training University, Tehran

2012 to 2014

SELECTED PUBLICATIONS

- Torkashvand, M., Koylu, C., & Kasakoff, A. B. (in review). Geocoding Birthplaces in Crowd-Sourced Family Trees: Temporal Toponym Resolution with Hierarchical Paths and a Human-in-the-Loop Large Language Model.
- Koylu, C., Kasakoff, A. B., & Torkashvand, M. (2025). US migration from 1850 to 1920: A comparison of family trees with linked census data. *Historical Methods*, 1 to 5.
- Koylu, C., Kasakoff, A. B., & Torkashvand, M. (2025). Settlement and the intergenerational dispersion of kin as a spatial process in the nineteenth-century US. *The History of the Family*, 706 to 729.
- Koylu, C., & Torkashvand, M. (2023). The effect of disruptive events on spatial and social interactions: Structural changes in pre- and post-COVID-19 pandemic networks. *GIScience 2023 Workshop on Disruptive Movement Analysis*.
- Torkashvand, M., Neshat, A., Javadi, S., & Pradhan, B. (2021). New hybrid evolutionary algorithm for optimizing index-based groundwater vulnerability assessment. *Journal of Hydrology*, 598, 126446.

SELECTED PROJECTS

- **Road Network Analysis and Routing** (Geospatial Programming coursework): Built programmatic routing and network-analysis workflows on road network data in ArcGIS Pro using the Network Analyst extension and Arcpy, applying solver-based approaches for route and origin-destination problems.
- **Meta-Coupled World Flows Modeling** (I-GUIDE Summer School, UCAR, 2023): Collaborated in a multidisciplinary team on an intensive short-term project examining the environmental and socioeconomic interconnection of world regions, modeling flows of goods, services, people, and capital as networks in R; presented the work as invited speaker at the [I-GUIDE Virtual Center](#), "Spatiotemporal Dynamics of a Meta-Coupled World" (March 2024).

CERTIFICATIONS AND AWARDS

- **Certifications:** Esri Spatial Data Science: The New Frontier in Analytics (2025); Interdisciplinary Graduate Certificate in Informatics, University of Iowa (2025); Artificial Intelligence Proficiency Levels 1 and 3, Iowa Initiative for AI (2025).
- **Selected Awards:** Ballard and Seashore Dissertation Fellowship, University of Iowa (\$11,653, Fall 2026); Graduate College Post-Comprehensive Research Fellowship (\$11,375, Spring 2026); Digital Scholarship and Publishing Summer Fellowship (\$5,500, Summer 2025); Best Student Paper Award, CaGIS + UCGIS Symposium (2024); I-GUIDE Summer School (selective geospatial data-science program, 2023).